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Duration	4 mins 44 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It is slower than linear search for small lists
- c. It does not work with negative numbers
- d. It can only be applied to large lists

Question **2**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Hash search
- b. Sequential search
- c. Binary search ✓
- d. Linear search

Question **3**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

- a. Selection
- b. Bubble ✓
- c. Insertion
- d. Complexity

Question **4**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all

b. When the item is somewhere in the middle of the array ✓

c. Item is the last element in the array or item is not there at all

d. When the item is the last element in the array

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array

b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

c. There must be a mechanism to access middle element directly

d. Binary search algorithm is not efficient when the data elements more than 1500 ✓

Question **6**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It can be applied only if the list is sorted ✓

b. It always starts from the beginning of the list

c. It compares elements sequentially

d. It works on unsorted lists

Question **7**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search has a lower time complexity on large, sorted lists ✓

b. Binary search works on unsorted lists

c. Binary search does not require dividing the list

d. Binary search can find multiple instances of the target element

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Sorting the list before searching
- b. Checking each element sequentially ✓
- c. Skipping every second element
- d. Dividing the list in half repeatedly

Question **9**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and unsorted ✓
- b. When the list is small and sorted
- c. When the list is large and unsorted
- d. When the list is large and sorted

Question **10**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. log n
- b. n ✓
- c. n/2
- d. 1

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the left sublist ✓
- b. At the beginning of the list
- c. In the entire list
- d. In the right sublist

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues from the beginning
c. The list is sorted
d. The search continues in the left or right sublist ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By dividing the list into halves
b. By comparing each element in the list sequentially ✓
c. By sorting the list first
d. By using a hash function

Question **14**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Divide the list into two equal parts
b. Compare the target element with the first element in the list
c. Sort the list
d. Compare the target element with the middle element in the list ✓

Question **15**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)

b. Hash search
c. Linear search ✓
d. Binary search